# Supplemental Text

**What Are Phases? Geographic Structure, Neutral Copying, and the Search for Social Boundaries in the Lower Mississippi Valley**

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*Supplement to the main text. References to numbered figures and tables without an "S" (for example, Figure 3, Table 2) point to the main article; Figures S1 and S2 and Tables S1 to S8 are in this supplement.*

## Statistical conventions

The conventions of the main text apply. Posterior quantities are medians with 90 percent equal-tailed intervals unless stated otherwise, each assemblage's class proportions are drawn from a Dirichlet posterior with a Dirichlet(1/2) prior, every Stan fit reports its sampler diagnostics, and signed quantities are stated with their direction in words. Three sections carried over from the earlier version of this study, on the minimum sample size, on the recovery of the phases, and on the second case, report 95 percent intervals and 95 percent simulation ranges and say so where they do.

## How this supplement is organized

The sections follow the order of the main text. The first three support the Data section: the minimum sample size of 75 decorated sherds, the assemblage coordinates and their datum, and the radiocarbon chronology. The next two support the first question, why culture historians saw phases in these data: how the clustering that recovers the phases was run and how the treatment of zero counts changes its answer, and the settlement record along the rivers and across the phase boundaries. The section that follows carries what the models of the main text require the supplement to hold: the defense of the spatial field model's priors, sampler diagnostics, the barrier model under six prior-by-range settings, the rim priors, and the cell-by-cell outcomes behind the neutral comparisons. A short section places Parkin in its settlement record, and the second case, on Mainfort's (2003a) table, follows it. A closing section points to the public code repository and the `phasecheck` tool. Each cross-reference in the main text names the material it refers to. The analyses of the earlier version of this study, which the restructured main text no longer carries, are not reproduced here; their scripts and findings files remain in the public repository.

## The minimum sample size, and why it is 75

This section explains why assemblages need at least 75 decorated sherds to enter the analysis. Cultural $F_{ST}$ computed directly from counts is biased upward by sampling. Two assemblages drawn from the same population of pots differ because each is a finite sample, and the spurious variance between them scales

roughly as $1/n$. To find the count at which that artifact becomes small relative to the differences we interpret, we drew pairs of multinomial samples of size $n$ from the basin's pooled class profile, 4,000 pairs at each size, and computed $F_{ST}$ between them. The two share one profile by construction, so all of the $F_{ST}$ between them is sampling (`analyses/82_minimum_sample_size.py`). The reference is the observed between-cluster $F_{ST}$ at five clusters, 0.0307 (Table S1).

**Table S1.** Spurious $F_{ST}$ between two multinomial samples drawn from the basin's pooled class profile at each sample size, 4,000 pairs per size (`analyses/82_minimum_sample_size.py`); the reference is the observed five-cluster $F_{ST}$ of 0.0307.

| decorated sherds per assemblage | spurious $F_{ST}$, median | 95th percentile | median as a share of the observed |
|---|---|---|---|
| 25 | 0.0146 | 0.0630 | 48% |
| 50 | 0.0070 | 0.0305 | 23% |
| **75** | 0.0044 | 0.0195 | 14% |
| 100 | 0.0034 | 0.0156 | 11% |
| 150 | 0.0023 | 0.0105 | 7% |
| 200 | 0.0016 | 0.0074 | 5% |
| 300 | 0.0011 | 0.0052 | 4% |
| 500 | 0.0006 | 0.0030 | 2% |
| 1000 | 0.0003 | 0.0015 | 1% |

At 75 sherds, the median artifact is about a seventh of the observed five-cluster differentiation, and its 95th percentile (0.0195) lies below it, whereas at 50 sherds, the 95th percentile (0.0305) is nearly equal to it. The five-cluster value is the reference because it is the finest division analyzed cluster by cluster (the earlier version's analyses 72 and 85), where clusters hold the fewest sherds and a sampling artifact would matter most. Against the two-cluster value of 0.0062, the artifact is proportionally larger at every sample size, which is why the copying model is always sampled at the real sherd counts: it then carries the same artifact as the data. Above 75, the bias shrinks slowly, by about three percentage points between 75 and 100, while assemblages are lost quickly: a minimum of 100 would remove three more basin assemblages (Davis, Irby, and Grant). The analytic expectation for two equal samples from one profile is close to $1/n$. Mainfort (2003) kept sites with 800 or more sherds in all. Decorated sherds are 13 percent of his table, so his rule corresponds to about 100 decorated sherds. The copying model is sampled at the observed sherd counts and carries the same bias, so the model comparison is not distorted by it. The minimum protects the values for single small clusters, where a value computed on a few dozen sherds would be mostly artifact.

## Coordinates and datum

Assemblage coordinates come from `data/raw/mainfort-pfg-cplXY.txt`, with published corrections applied on reading (`data/raw/coordinate_corrections.csv`): Belle Meade and Beck are moved to author-supplied coordinates inside the sections Phillips, Ford, and Griffin record, and Starkley and Davis (13-N-5) to the survey's own coordinates, which fall inside their recorded sections (for Davis, SE1/4 SE1/4 of Section 14, T3N R5E, confirmed against the Bureau of Land Management's cadastral survey layer). Cummins (11-O-4) is moved 22.7 km, from a file coordinate in quadrangle 10-N that belongs to a different site joined on the name (Arkansas record 3PO243, a Dalton site), to the coordinate the From LMV Project sheet of `data/raw/mainfort-spatial-data.xlsx` gives for it, which lies inside quadrangle 11-O as its designation requires and 0.76 km from the site numbered 4 that Lipo (2001:Figure 2.6) plots there. This correction is provisional until the section Phillips, Ford, and Griffin give for 11-O-4 is supplied. The survey's coordinates were recorded in the 1927 North American Datum. Reading them in the 1983 datum instead places each point about 209 m south of where the survey placed it, so we convert them. We convert only coordinates that can be derived from the survey: an assemblage coordinate is converted when it lies within 10 m of the survey coordinate for that site read in the 1983 datum. That applies to 19 assemblages, 15 of them in the basin set, each moved 209 m. Eleven further coordinates lie within 100 m of that position but outside the tolerance (14 to 54 m for the eight in the basin set), which means they were derived another way. Eight of them are Mainfort's own points, and all eleven are left as recorded. Settlement-table rows whose source is the survey (367 rows) are converted the same way. The conversion leaves the $k$-means divisions at two, three, five, and six clusters and their direct $F_{ST}$ unchanged, and changes the division at four clusters (the four-cluster value moves from 0.0194 to 0.0186). Every result reported uses the converted coordinates (`scripts/measure_datum.py`, `output/findings/datum_conversion.md`).

## Radiocarbon chronology

This section gives the radiocarbon evidence for when the basin was occupied and shows why the seriation order cannot be interpreted as a chronology. We calibrated the Mainfort (2001) radiocarbon compilation against IntCal20 (Reimer et al. 2020). Forty determinations come from seven sites in and near the St. Francis basin, nineteen of them from Parkin. The summed probability of the basin dates has a median of AD 1428 (Parkin alone, AD 1483). It places 74 percent of the probability between AD 1241 and AD 1641, spread evenly over those four centuries (18 or 19 percent each), with 16 percent earlier and 10 percent in the century after (Figure S1; `output/chronology_14c.md`). It does not fall at the contact interval. The century after AD 1541 holds 19 percent, the same as the century before. In all, 29 percent of the basin probability postdates AD 1541 and 18 percent postdates AD 1600. These shares are computed on calendar years AD 1000 to 1749, which assumes that no deposit postdates 1750.

Calibrated without that bound, 8 percent of the probability falls after AD 1741, the shares after AD 1541 and AD 1600 become 31 and 21 percent, and the centuries before and after AD 1541 remain equal, at 17 percent each. This is consistent with occupation continuing through the De Soto entrada and declining afterward, although the shape of the calibration curve affects the shape of any summed distribution (Surovell and Brantingham 2007), and the latest tail is partly a product of determinations near the post-1650 plateau.

To test whether the seriation axis tracks calendar time, we pooled the dates of the four proveniences matching our assemblages (Kent Place, Parkin, Neeley's Ferry, and Clay Hill; 26 dates) and correlated their median calibrated ages with position on the first correspondence-analysis axis. With pooled calibrated medians, the rank correlation is 0.40 in magnitude, with Clay Hill out of order, and it favors orienting the axis the way the analyses do. With one-sigma calibrated midpoints, it is zero. Four points under two reasonable conventions that disagree cannot establish which end of the axis is later.

House (1993:23) reports the Kent and Clay Hill determinations with their contexts. At Kent, one date comes from a house floor (Feature 6, SMU-2102, 600 ± 115 BP) and the other from a post intrusive into it (SMU-2103, 310 ± 90 BP), and an archaeomagnetic date on another feature (Feature 1) falls at AD 1345 to 1380. Pooling the intrusive post with the floor makes the Kent Place median (AD 1466) later than the floor itself. Correcting it would not orient the axis either, because Kent Place, then among the earliest dated sites, and Clay Hill, the latest, sit at the same end of the axis. Hollywood, one of the analyzed assemblages but not in the radiocarbon file, has two calibrated dates reported by Haley (2014) from Johnson and colleagues' excavations (Johnson et al. 2000:48-49, as cited there): central intercepts of AD 1310, 1355, and 1385 (one-sigma AD 1285 to 1420) for the earliest features, and AD 1405 (1300 to 1435) for the top of Mound A. The uncalibrated ages are not reported, so we have not recalibrated them against IntCal20 or pooled them with the other dated assemblages. Hollywood too lies at the same end of the axis as Kent Place and Clay Hill (ranks 6, 4, and 7 of 28, with Parkin at 21).

The axis follows latitude, with a rank correlation of +0.77 as oriented, and latitude and longitude together account for 63 percent of its variance (`analyses/87_axis_geography.py`). It is largely a south-to-north gradient in composition, which is what the main text's argument would lead one to expect, and it is not a reliable proxy for time. The analyses therefore orient the axis in the direction the medians weakly favor, and report every direction-dependent result both ways.

The basin chronology is stable under an alternative compilation: adding the net-new Mississippian dates from the lower-valley radiocarbon database compiled by Jeffrey Alvey (personal communication, 2022), twelve determinations concentrated at Hazel and Upper Nodena, shifts the summed-probability median by eight years, to AD 1436, and leaves the contact decline in place (26 percent of the probability after AD 1541 and 16 percent after AD 1600). Recent AMS dating

of the Parkin platform-mound post, interpreted as the base of the cross the De Soto expedition raised at Casqui, places it at cal AD 1445 to 1650 (Mitchem et al. 2017), consistent with the latest occupation inferred here. The individual determinations, with laboratory numbers, ages, calibrated medians, and sources, are listed in `radiocarbon_dates_used.csv`.

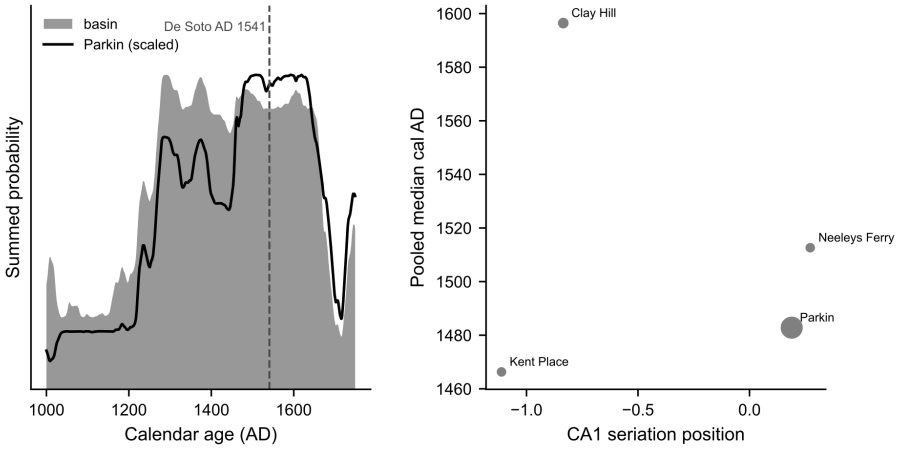



**Figure S1.** Radiocarbon chronology of the St. Francis basin (Mainfort determinations calibrated against IntCal20). Left: summed probability for the 40 determinations from seven sites in and near the basin (filled) and for Parkin alone (line, scaled), with the De Soto entrada (AD 1541) marked. The summed probability is level from the mid-thirteenth to the mid-seventeenth century, including across AD 1541, and falls after that. Right: position on the first correspondence-analysis axis against pooled median calibrated age for the four proveniences matching the analyzed assemblages (point size proportional to the number of dates). The axis is oriented as in the analyses; the rank correlation is +0.40, with the late-dated Clay Hill out of order, and four points do not establish the axis's direction.

## What recovers the phases, and the problem of zeros

The main text asks whether the site map or the assemblages recover the phase assignments. This section describes how assemblages were clustered on location and composition together, and one choice, the treatment of zero counts, that changes the answer.

Assemblages are clustered in a combined space: site coordinates in kilometers, centered and scaled so that the mean squared pairwise distance is one, joined to decorated-class composition scaled the same way and multiplied by the square root of a weight $w$. At $w = 0$, the representation is the site map alone. As $w$ grows, it becomes assemblage composition alone. We compare each clustering with the phase assignments, taken from Phillips's (1970) phase areas as mapped by Lipo (2001:Figure 2.6), using the adjusted Rand index. We ran

three composition transforms, three clustering methods ($k$-means over 40 seeds, each the best of 500 restarts; Ward linkage; and average linkage), and nine weights from the map alone upward, plus composition alone. The full grid is in `output/findings/phase_recovery.csv`. Two of the 28 assemblages, Cramor Place (Kent) and Holden Lake (Parkin), each 1.0 km outside its phase area, lie inside no phase area and are assigned to the nearest area, a geographic rule, so every value quoted in the main text is computed on the 26 assemblages inside a phase area, apart from the sensitivity case below, which scores all 28. The agreement reached by the site map alone depends on the clustering method. It is 0.480 by $k$-means and 0.740 by Ward linkage and by average linkage, whichever composition transform is used, since at $w = 0$ the transform does not enter (`output/findings/phase_recovery.md`). The $k$-means value is no better than the 0.485 that divisions with the phases' group sizes around random centers reach. With the chi-square transform, $k$-means reaches 0.740 from a quarter of the coordinates' weight up to equal weight, with the same two assemblages misplaced, and adding composition does not raise the Ward value. The difference lies in which assemblages each method places with another phase. Ward and average linkage misplace two, Big Eddy and Castile Landing, both in Phillips's Parkin area and both grouped with Kent. $K$-means misplaces those two and four more, Beck, Belle Meade, Commerce, and Hollywood, which it groups with Walls. With composition at half the coordinates' weight and the chi-square transform, all three methods misplace only Big Eddy and Castile Landing. Composition alone misplaces six of the 26 by $k$-means and by Ward linkage and thirteen by average linkage. These lists are from the chi-square transform and, for $k$-means, the first seed.

We also asked of each assemblage which phase its pottery fits best (`analyses/96_assemblage_phase_fit.py`, `output/findings/assemblage_phase_fit.md`). The measure is the chi-square distance from the assemblage's profile to the pooled profile of each phase, with the assemblage left out of its own phase's pool. Each profile is drawn 4,000 times from Dirichlet(counts + 1/2), and the posterior probability that an assemblage fits its own phase is the share of draws in which its own phase is the closest. That probability is above 0.75 for 22 of the 28 assemblages and below 0.25 for four: Big Eddy, Holden Lake, and Turnbow, which are closer to Kent, and Belle Meade, which is closer to Parkin. For Castile Landing it is 0.60 and for Mound Place 0.40: Castile Landing is nearly as close to Kent (distance 0.571) as to its own Parkin (0.555), and Mound Place is slightly closer to Kent and to Parkin (0.496 each) than to its own Walls (0.538). Two assemblages lie nearer a member of another phase than any other member of their own: Castile Landing (12 km from the nearest Kent assemblage, 16 km from the nearest Parkin assemblage) and Clay Hill.

We also asked whether assemblages made by copying with no groups would sort into the phases, using the copying model as the earlier version of this study calibrated it (`analyses/79_phases_under_drift.py`, `output/findings/phases_under_drift.md`). We ran the copying model on the real site map 300 times, sampled its type frequencies at the real sherd counts,

and clustered each simulated record the same way as the real one. Under Ward linkage, with a quarter weight on assemblage composition, the simulated records match the phases at a median of 0.740, the real record's value, and 68 percent reach it. At half weight, 36 percent do. At equal weight, the simulated median falls to 0.487, and 6 percent reach the real 0.740. The decline is not steady: at twice the coordinates' weight, the real record's own agreement falls to 0.454, and 46 percent of simulated records reach it. Based on type frequencies alone, the real record's agreement of 0.358 lies near the top of the simulated range (median 0.098, 95 percent range -0.045 to 0.416), and 6 percent of simulated records reach it. Part of the real record's advantage on composition is expected by construction, because Phillips drew the phases partly from these type frequencies, so the real pottery was sorted to fit them and simulated pottery was not. $K$-means gives the same result at equal weight and on composition alone (6 and 5 percent). At a quarter weight, it shows a wider difference (13 percent), because in the real record the pottery changes the division $k$-means makes of the site map, and in most simulated records it does not.

As a sensitivity case, we counted Big Eddy and Castile Landing as Kent and reran the recovery. Ward linkage on the site map alone then reproduces the scheme exactly, as does $k$-means on the map and composition together, and $k$-means on the map alone reaches 0.666 against 0.509 (all 28 assemblages scored). The primary analysis keeps Phillips's assignments.

Class proportions are compositional data, and the centered log-ratio transform is the usual choice. It cannot be applied here without a choice that determines the result. Of the 280 cells in the basin matrix, 105 are zero, and log (0) is undefined. Replacing zeros with a fraction of the smallest positive proportion lets that fraction set how strongly absence counts, and agreement with the phases follows it (Table S2):

**Table S2.** Agreement between the recovered clustering and the phases (adjusted Rand index) under each convention for replacing zeros before the centered log-ratio transform.

| zero convention | adjusted Rand index |
|---|---|
| pseudocount of 0.5 added to counts | 0.522 |
| floor at 2.0 times the smallest positive proportion | 0.740 |
| floor at 0.5 times the smallest positive proportion | 0.740 |
| floor at 0.1 times the smallest positive proportion | 0.863 |

All four use the same data, geography, and weight ($w = 0.5$). Only the zero convention differs, and it moves the answer by about a third of the index's range. We report the chi-square transform in the main text because it needs no such choice, and we use a pseudocount wherever the log-ratio transform appears. The result is also a caution about the phases themselves: a classification sensitive to

which types are absent, applied to assemblages whose sherd counts differ more than twentyfold, will interpret sampling failure as cultural difference.

## Settlement and waterways

The main text reports that settlement clumps follow the larger rivers (Figure 3B). This section asks whether nearness to water accounts for the clumping, and gives the measurements (`analyses/81_settlement_and_rivers.py`, `output/findings/settlement_and_rivers.md`). The sites are the 81 with a late pre-contact component in the mapped area, and the comparison is with 6,000 points placed uniformly at random in the same area. Clumping is measured by the Clark-Evans ratio, the mean distance to the nearest neighbor divided by its expectation for randomly placed points, which is 1 for a random pattern and less than 1 for a clumped one. The sites' ratio is 0.67.

The main text also reports that the phase boundaries run through sparsely settled country (`analyses/80_boundaries_in_settlement_gaps.py`, `output/findings/boundaries_in_settlement_gaps.md`; main-text Figure 3B). The comparison uses the 53 late-period sites that lie more than a kilometer from any analyzed assemblage, and the reconstructed boundaries described in the main text. Measured at a 5 km scale, 94 percent of same-size divisions around random centers and 100 percent of compact ones run through denser late-period settlement than the phase boundaries do, and the result is similar at the other two scales tested, 98 and 100 percent at 3 km and 89 and 86 percent at 8 km.

Any waterway mattered on a floodplain that the Mississippi regularly flooded, so we counted streams of every size. The stream network is HydroRIVERS (Lehner and Grill 2013), with reaches added from the largest flow order to the smallest it holds. For each level, Table S3 gives the distance from sites and from random points to the nearest stream, and the Clark-Evans ratio of random points placed at the sites' own distances from the streams. If nearness to water produced the clumping, those points would be as clumped as the sites.

**Table S3.** Late-period sites and waterways. Median distance to the nearest stream for the 81 late-period sites and for 6,000 random points, as smaller streams are counted, and the Clark-Evans ratio of random points placed at the sites' distances from the streams (median of 300 placements, with 2.5th to 97.5th percentiles). The sites' own ratio is 0.67.

| streams counted | channel length (km) | smallest mean discharge ($m^3/s$) | sites, median km (quartiles) | random points, median km (quartiles) | sites within 1 km | random points within 1 km | Clark-Evans ratio of points at the sites' distances |
|---|---|---|---|---|---|---|---|
| the Mississippi | 168 | 15,242 | 10.15 (5.87 to 33.48) | 19.66 (8.37 to 35.69) | 1% | 3% | 0.90 (0.79 to 1.01) |
| large rivers | 370 | 104 | 4.83 (1.53 to 7.94) | 9.50 (4.07 to 16.04) | 17% | 7% | 0.83 (0.72 to 0.93) |
| and mid-sized streams | 847 | 10.2 | 2.00 (0.69 to 5.87) | 3.84 (1.61 to 7.04) | 31% | 16% | 1.00 (0.87 to 1.13) |
| and small streams | 1,909 | 1.0 | 1.30 (0.58 to 2.01) | 1.63 (0.75 to 2.85) | 41% | 32% | 1.03 (0.89 to 1.16) |
| every reach | 3,235 | 0.1 | 0.76 (0.36 to 1.37) | 0.93 (0.43 to 1.52) | 58% | 53% | 1.04 (0.90 to 1.14) |

Sites favor the larger rivers. They lie about half as far from the large and mid-sized streams as random points do. That preference weakens as smaller streams are counted, because small streams are everywhere on this floodplain, and with every reach included sites are barely nearer water than random points are (0.76 against 0.93 km). Nearness to water does not produce the clumping at any level. Points placed at the sites' distances from the large rivers are mildly clumped (0.83), still well short of the sites' 0.67, and at every finer level they are randomly spaced.

Two limits apply. The network is modern, and channels have moved since these sites were occupied. HydroRIVERS is also derived from an elevation grid of about 500 m, so on flat alluvial ground its small reaches are computed flow paths rather than surveyed channels, and it holds no oxbow lakes, bayous, or sloughs. Many of the sites carry the names of such lakes, which suggests that settlement followed abandoned channels as much as active ones, but we have no layer with

which to test that.

## Supporting material for the models of the main text

The analyses of the main text are those of the public repository's `output/findings/` directory, each named in Table 1 of the main text. This section carries what those analyses require the supplement to hold: the defense of the spatial field model's priors, sampler diagnostics, the prior-by-range comparison for the barrier model, the defense of the rim priors, and the cell-by-cell outcomes behind the neutral comparisons.

**Intervals.** This section reports posterior medians with 90 percent equal-tailed intervals (the 5th to 95th percentiles), matching the main text and the findings files.

**The spatial field model's priors (rule 20).** The three parts of the defense were run on the basin fit under the inherited Normal(0, 2) class-intercept prior (`a0_basin_refit.md`, `a0_predictive_summaries.md`), before the Normal(0, 10) default was adopted; the rerun under the default prior is an open item. Direction: the half-Normal(0, 2) prior on the field scale sigma and the site-level scale tau admits large variance of both kinds, and the 1/rho prior on the range puts half its mass below 12.2 km, the geometric mean of 0.5 and 300 km. Prior predictive on the observable: 200 records drawn from the priors alone at the real geometry and sherd totals give a mean within-assemblage Gini-Simpson diversity of 0.41 (5th to 95th percentile 0.22 to 0.67) against the observed 0.56, a mean class richness of 6.2 (3.9 to 8.9) against 6.25, a pooled Gini-Simpson diversity of 0.75 (0.35 to 0.84) against 0.59, and a plug-in two-cluster $F_{ST}$ of 0.028 (0.002 to 0.176) against 0.0062, so the priors do not already resemble the record on any of the four summaries. The posterior predictive medians on the same summaries are 0.565, 6.5, 0.592 and 0.0064. Prior sensitivity: under half-Normal(0, 0.5) on sigma and tau, which leans toward little variance of either kind, the spatial share is 0.95 (0.88 to 1.00) against 0.99 (0.96 to 1.00) under the default, and under a uniform prior on rho over 0.5 to 300 km, which leans toward long ranges, it is 0.99 (0.96 to 1.00); the between-site contrast standard deviations at 4.3, 36.6 and 79.1 km move by at most 0.8 logits. The three class-intercept priors are compared in the main text (Figure 4). Recovery against the prior: on records simulated from the model at the real geometry and sherd totals with a spatial share of 0.2 and 0.9 crossed with a range of 5 and 60 km, the 90 percent interval covers the planted share in all four cases (Table S4), and the range and scale are recovered only at a share of 0.9, which is the identification limit the main text states.

**Table S4.** Recovery of the spatial field model's quantities on four records simulated from the model at the 28 sites (`a0_basin_refit.md`, section 2c): planted value, posterior median and 90 percent interval, under the inherited Normal(0, 2) class-intercept prior.

| record | quantity | planted | median | 5% | 95% |
|---|---|---|---|---|---|
| share 0.2, range 5 km | spatial share | 0.200 | 0.361 | 0.004 | 0.990 |
| share 0.2, range 5 km | rho (km) | 5.0 | 1.9 | 0.6 | 131.8 |
| share 0.2, range 60 km | spatial share | 0.200 | 0.155 | 0.001 | 0.961 |
| share 0.2, range 60 km | rho (km) | 60.0 | 2.5 | 0.6 | 189.6 |
| share 0.9, range 5 km | spatial share | 0.900 | 0.923 | 0.529 | 0.999 |
| share 0.9, range 5 km | rho (km) | 5.0 | 4.2 | 2.5 | 8.4 |
| share 0.9, range 60 km | spatial share | 0.900 | 0.900 | 0.787 | 0.963 |
| share 0.9, range 60 km | rho (km) | 60.0 | 67.2 | 23.0 | 224.8 |

**The latent-unit model under the corrected intercept prior.** Three A1 fits were repeated with the Normal(0, 10) class-intercept prior (`a1_prior_check.md`): on the record simulated with no units the posterior median of $\sigma_g$ is 0.58 (0.06 to 1.47) against 0.52 (0.05 to 1.39) under Normal(0, 2), and on two records planted at 0.5 it is 0.26 and 0.27 against 0.27 and 0.27. All three fits fail the sampler diagnostics under both priors.

**Sampler diagnostics (rule 16 of the project's verification regime).** Every Stan fit reported in the main text carries R-hat, bulk and tail effective sample size, divergent transitions, treedepth saturation and the energy Bayesian fraction of missing information. The three A0 fits that differ in the class-intercept prior (`a0_intercept_prior.md`): Normal(0, 2), R-hat max 1.0034 (lp___) | bulk ESS min 608 (tau[1]) | tail ESS min 1249 (tau[1]) | divergences 0/6000 (0.00%) | treedepth at max 12: 0/6000 (0.0%) | E-BFMI min 0.676 | stuck 0; Normal(0, 10), R-hat max 1.0059 (lp**) | bulk ESS min 874 (tau[1]) | tail ESS min 1666 (tau[1]) | divergences 0/6000 (0.00%) | treedepth at max 12: 0/6000 (0.0%) | E-BFMI min 0.672 | stuck 0; Normal(pooled log-odds, 2), R-hat max 1.0050 (lp**) | bulk ESS min 1053 (tau[1]) | tail ESS min 1928 (tau[1]) | divergences 0/6000 (0.00%) | treedepth at max 12: 0/6000 (0.0%)

| E-BFMI min 0.720 | stuck 0. The A1 fit to the record simulated with no units (`a1_validation.md`): R-hat max 1.0244 (comembership[27,25]) | bulk ESS min 277 (comembership[25,21]) | tail ESS min 450 (comembership[14,11]) | divergences 0/12000 (0.00%) | treedepth at max 13: 0/12000 (0.0%) | E-BFMI min 0.632 | stuck 0; this fit fails the diagnostics on R-hat and bulk effective sample size and is reported as a failed fit. The A3 equivalence fits (`a3_validation.md`): A0 on the same record, R-hat max 1.0040 (alpha[5]) | bulk ESS min 1135 (rho) | tail ESS min 1708 (rho) | divergences 0/6000 (0.00%) | treedepth at max 12: 0/6000 (0.0%) | E-BFMI min 0.742 | stuck 0; A3 with the barrier fixed at zero, R-hat max 1.0053 (rho) | bulk ESS min 973 (rho) | tail ESS min 1500 (rho) | divergences 0/6000 (0.00%) | treedepth at max 12: 0/6000 (0.0%) | E-BFMI min 0.728 | stuck 0. For the A1 recovery fits (14) and the A3 recovery and pinned-range fits (39 per prior-by-range setting), the committed tables record the diagnostic outcome and, for a failure, which diagnostics failed; the numeric values were printed to the console during fitting and were not saved. Twelve of the 14 A1 recovery fits fail (eleven on R-hat or effective sample size of the co-membership probabilities, one on divergent transitions); all 39 A3 free-range fits pass under both intercept priors; with the range pinned at 12 km under the Normal(0, 10) prior, 37 of 39 pass. The export of per-fit numeric diagnostics from the fit cache is an open item. The rim posteriors are direct conjugate draws (4,000 draws, two seeds), for which sampler diagnostics do not apply; their Monte Carlo errors are given beside each median in `rim_r1.md`.

**The barrier model under six prior-by-range settings (Table S5).** Figure 7B of the main text shows the free-range fits under the Normal(0, 10) class-intercept prior. Table S5 gives the range of diagnostic-passing posterior medians of the barrier parameter omega for every combination of range setting and intercept prior, over the own-generator records (`a3_prior_comparison.md`). The prior on omega is half-Normal(0, 2), median 1.35. The posteriors move with the setting, and under the inherited Normal(0, 2) prior with the range pinned they sit well above the prior median at every planted value; under no setting do the planted values separate.

**Table S5.** Posterior medians of the barrier parameter omega on records simulated from the model's own generator at the 28 sites, by range setting and class-intercept prior: the range of diagnostic-passing medians over the records at each planted value, with the number of passing fits. Diagnostics rows give the number of fits passing of 39.

| range | planted omega | Normal(0, 2) intercept prior: posterior medians | Normal(0, 10) intercept prior: posterior medians |
|---|---|---|---|
| free | 0 | 0.47 to 1.16 (n 6) | 0.48 to 1.21 (n 6) |
| free | 0.25 | 0.44 to 1.14 (n 6) | 0.43 to 1.08 (n 6) |
| free | 0.5 | 0.64 to 2.05 (n 6) | 0.61 to 1.86 (n 6) |
| free | 1 | 0.76 to 2.59 (n 6) | 0.70 to 2.28 (n 6) |

| range | planted omega | Normal(0, 2) intercept prior: posterior medians | Normal(0, 10) intercept prior: posterior medians |
|---|---|---|---|
| free | diagnostics | 39 of 39 pass (at the first sampler setting in 27) | 39 of 39 pass (at the first sampler setting in 30) |
| 6 km | 0 | 2.83 to 3.49 (n 6) | 0.69 to 1.58 (n 6) |
| 6 km | 0.25 | 2.58 to 3.35 (n 6) | 0.81 to 1.57 (n 6) |
| 6 km | 0.5 | 3.12 to 3.72 (n 6) | 1.42 to 2.04 (n 6) |
| 6 km | 1 | 3.01 to 4.20 (n 6) | 1.05 to 2.50 (n 6) |
| 6 km | diagnostics | 39 of 39 pass (at the first sampler setting in 34) | 39 of 39 pass (at the first sampler setting in 31) |
| 12 km | 0 | 2.74 to 3.40 (n 6) | 1.03 to 1.51 (n 5) |
| 12 km | 0.25 | 2.34 to 3.25 (n 6) | 0.57 to 1.44 (n 5) |
| 12 km | 0.5 | 2.91 to 3.58 (n 6) | 1.22 to 1.96 (n 6) |
| 12 km | 1 | 2.93 to 4.19 (n 6) | 0.81 to 2.38 (n 6) |
| 12 km | diagnostics | 39 of 39 pass (at the first sampler setting in 30) | 37 of 39 pass (at the first sampler setting in 22); fail: own_G1_om0_r3, own_L_om0.25_r3 |

**The rim priors (rule 20).** Each site frequency of a rim attribute carries a Beta(1/2, 1/2) prior and each composition a Dirichlet(1/2) prior. The direction: one pseudo-observation in total, weight $1/(N + 1)$ on the posterior mean (0.036 at 27 rims, 0.0036 at 277), toward one half. The prior predictive on the observable: under Beta-Binomial(N, 1/2, 1/2) at each site's rim count, the 5th to 95th percentile interval of the predicted count spans 99 to 100 percent of N at every site and contains all 57 observed counts for the three rim attributes, so the prior concentrates the predicted count nowhere and does not already resemble the data. Sensitivity: Beta(1, 1) and Dirichlet(1) medians are reported beside every posterior in `rim_r1.md` sections B and C and move the latitude and pair-association medians by at most a few hundredths. Recovery: 20 replicates at three true configurations on the 19 sites' rim totals, 1,000 draws per fit, 90 percent intervals, recover the sign of the latitude and pair-distance correlations in every replicate; at 20 replicates a coverage proportion has binomial standard deviation 0.067 at a true rate of 0.9, so the program resolves gross failure, not the nominal rate (`rim_r1.md` section F).

**The two-cluster F_ST against the neutral model, by cell.** The observed cultural F_ST between the two spatial clusters the site layout supports, 0.0062, is compared against the neutral band in 30 cells: the ten C1 cells (accumulation span, two operating points; inside the band in 7), the ten C2 cells (span tied to collection size; inside in 6), the five C3 cells at the primary operating point

(regional drift amplitude; inside in 3) and the five at the secondary (inside in 1), for 17 of 30 in total. The cells it lies above are those with the longest spans and the smallest drift amplitudes, which narrow the band from below. The per-cell values are in `per_type_generators_C1.md`, `_C2.md`, `_C3_primary.md` and `_C3_secondary.md`.

**Rare-class occupancy by cell (Table S6).** The number of the 28 sites at which each rare class is present, against the neutral median and 10th to 90th percentile band at each regional-drift amplitude at the primary operating point (`per_type_drift_sensitivity.md`). Only "outside" above the band is interpreted as more variation than neutral copying produces. All seven classes are covered together only at amplitudes 0.01 and 0.03 at the primary point and at no tested amplitude at the secondary.

**Table S6.** Sites present for each rare class in the record, and the neutral median [10th, 90th percentile] with the record's status, by regional-drift amplitude $F_W$ at the primary operating point (50 records per cell).

| class | record (sites of 28) | $F_W = 0$ | 0.01 | 0.03 | 0.1 | 0.3 |
|---|---|---|---|---|---|---|
| Fortune Noded | 15 | 19 [16, 23], outside | 20 [10, 25], inside | 12 [2, 25], inside | 9 [2, 26], inside | 4 [0, 21], inside |
| Ranch Incised | 22 | 23 [19, 25], inside | 22 [13, 26], inside | 20 [7, 26], inside | 8 [2, 27], inside | 6 [1, 24], inside |
| Walls Engraved | 14 | 18 [14, 21], inside | 16 [9, 22], inside | 14 [3, 24], inside | 8 [0, 25], inside | 3 [0, 28], inside |
| Wallace Incised | 7 | 8 [5, 14], inside | 7 [1, 16], inside | 5 [0, 21], inside | 1 [0, 7], inside | 0 [0, 6], outside |
| Rhodes Incised | 15 | 11 [7, 18], inside | 10 [1, 18], inside | 4 [0, 23], inside | 1 [0, 19], inside | 0 [0, 13], outside |
| Vernon Paul Applique | 10 | 7 [3, 12], inside | 4 [0, 14], inside | 2 [0, 16], inside | 0 [0, 11], inside | 0 [0, 3], outside |
| Hull Engraved | 8 | 4 [1, 6], outside | 0 [0, 11], inside | 0 [0, 8], inside | 0 [0, 3], outside | 0 [0, 3], outside |

## Parkin in the settlement record

The main text concludes that decorated frequencies do not establish a boundary in ceramic learning at the edge of the Parkin phase. Two descriptions place Parkin in its regional context without bearing on that boundary.

The settlement record provides further context. Parkin's principal mound is a flat-topped platform 6.5 m (21.3 ft) high on the site's western edge, overlooking the St. Francis River, with six smaller mounds near it. The site covers 17 acres (6.9 ha), is ditched on three sides, and its deposits rise 3 m above the floodplain (Morse 1981, 1990). Its mound is the second tallest of the 55 basin sites with a recorded height, behind Mound Cemetery (13-N-3) at 23 ft, and the tallest exceeds the second by a ratio of only 1.08 (Figure S2). Parkin's height comes from excavation reports, whereas the others come from a regional compilation without per-site sources, so the series mixes measurement sources. Mound heights form a graded series with no gap at the top. Maximum mound height is a limited proxy for political rank: it records the final size of one construction, not the organization that built it, and a graded series fits more than one form of ranking, so it describes the settlement record without testing for an administrative tier.

Regional evidence on the production and exchange of stone hoes adds context: autonomous workshops made Mill Creek hoes at the source and distributed them largely through successive exchanges, with elites at consuming centers gaining at most privileged access (Cobb 2000:75, 154, 198). That evidence is not basin-specific and cannot test Parkin's position locally.

## The second case: the tests on Mainfort's (2003a) table

The main text analyzes Mainfort's (2003) table and phases separately from the primary dataset, as a second case. His assignments follow Phillips's, with Smith's (1990) Horseshoe Lake phase split out and the phases of western Tennessee added (Mainfort 2003:176–177), and his table is not a larger sample once restricted to decorated classes. This section details that analysis. The tests are those of the earlier version of this study: whether the site map recovers his phases, and whether his phase lines separate the assemblages better than same-size divisions around random centers and compact divisions. The neutral copying model of the main text was not run on his table.

Mainfort's Table 1 gives counts of ten decorated types for 39 sites with 800 or more sherds in all. His types map onto our classes as follows: Barton Incised and Kent Incised join the class that also holds Mound Place Incised, and Old Town Red and Nodena Red and White join the painted class. He does not tally Wallace Incised or Hull Engraved, which are carried as zeros so the class list matches the main analysis. Coordinates come from his own site list where it has them (30 sites), from this project's assemblage file for eight (four of them with a project correction), and from the settlement compilation for Sweat, and each row of `data/processed/mainfort2003_sites.csv` names its source. The file is written by `scripts/build_mainfort2003_sites.py`. The analysis uses decorated types only and sites with at least 100 decorated sherds, stricter than the main analysis's 75 because the second case should not be the looser test. Nine of his 39 sites fall below that, and Tipton's one remaining site, Graves Lake, is not carried as a phase of one. The tests are those of `analyses/74_phase_partition_test.py` and

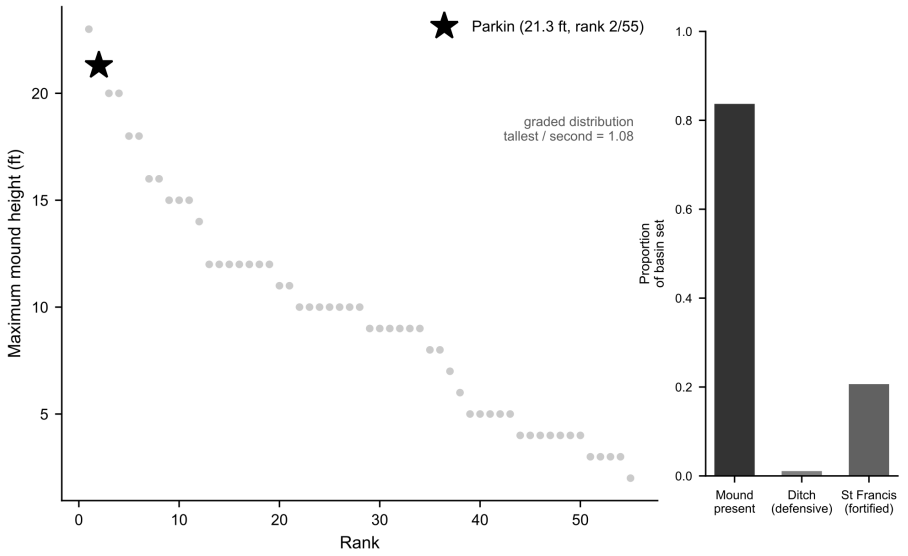



**Figure S2.** Settlement cross-check. Left: maximum mound height ranked across the basin, the only size attribute the regional compilation records for most sites, from non-uniform sources (mound area is recorded as zero for Parkin and is not used). Parkin's principal mound is 21.3 ft (6.5 m) after Morse (1981, 1990), second among the 55 basin sites with a recorded height, behind Mound Cemetery (13-N-3) at 23 ft. The compilation also gives Parkin 23 ft, so we substitute the excavated value. The distribution is graded, with no gap at the top (tallest-to-second ratio 1.08). Right: the proportion of sites with a mound, a defensive ditch, and St. Francis-type fortification across the basin set (n = 92).

`analyses/76_phase_recovery.py`, run by `analyses/83_mainfort_replication.py` under his labels. A second reader checked the transcription of his table against the published page. The one row whose cells do not sum to its printed total, Hatchie (818 against 814), is transcribed correctly, and no analysis uses the printed totals (Table S7).

**Table S7.** Recovery of Mainfort's (2003a) phases by clustering his sites at each weight on composition, adjusted Rand index with his phases, median over 40 starts (`analyses/83_mainfort_replication.py`).

| weight on composition | agreement with his phases (adjusted Rand index, median over 40 starts) |
|---|---|
| map alone | 0.536 |
| 0.1 | 0.536 |
| 0.25 | 0.509 |
| 0.5 | 0.664 |
| 1 | 0.483 |
| 2 | 0.483 |
| composition alone | 0.248 |
| same sizes, boundaries at random (400 draws) | 0.306 |

All 40 seeds return the same value at every weight.

Under his phases, the cultural $F_{ST}$ is 0.0470 (direct), with a posterior median of 0.0459 (95 percent credible interval 0.0413 to 0.0505) when each site's class proportions are drawn from Dirichlet(counts + 1/2). Same-size divisions around random centers give a posterior median of 0.0297 and compact divisions 0.0315, and the posterior probability that his lines separate the assemblages better is 0.96 and 0.94 (`analyses/86_partition_posterior.py`). The compactness search returned his division in none of 1,000 starts (Table S8).

**Table S8.** Sites the two schemes assign differently: the phase under each scheme, the ceramic distance from the site to each phase's pooled profile, and the distance in km to that phase.

| site | Mainfort 2003a | primary analysis | ceramic distance to his phase | km | ceramic distance to the primary-analysis phase | km |
|---|---|---|---|---|---|---|
| Beck | Horseshoe Lake | Kent | 0.333 | 15 | 0.242 | 14 |

| site | Mainfort 2003a | primary analysis | ceramic distance to his phase | km | ceramic distance to the primary-analysis phase | km |
|---|---|---|---|---|---|---|
| Belle Meade | Horseshoe Lake | Kent | 0.256 | 13 | 0.521 | 15 |
| Mound Place | Horseshoe Lake | Walls | 0.651 | 17 | 0.778 | 7 |
| Young | Horseshoe Lake | Walls | 0.467 | 13 | 0.443 | 15 |
| Carson Lake | Walls | Nodena | 1.039 | 71 | no other Nodena site at the minimum | |

Distances are chi-square distances from the site's profile to the pooled profile of the candidate phase with the site left out, and straight-line kilometers to that phase's center. His Parkin Punctated counts equal the survey's at 11 of the 21 shared sites and exceed them at ten (`output/findings/matrix_integrity.md`), so for those sites this is a second sampling of the same collections. Only the western Tennessee sites, from Tennessee Division of Archaeology collections, are material the survey never had. The copying model is not rerun on this table because it is calibrated to the main matrix, and recalibrating it would be a separate study.

## Code, data, and the phasecheck tool

All code and data are at https://github.com/clipo/phases, the code under the MIT License and the data under a Creative Commons Attribution 4.0 International license; a versioned archive will be deposited at https://doi.org/10.5281/zenodo.20796240 on publication. The repository holds the source tables (`data/raw/`), the processed matrix (`data/processed/`), the numbered analysis scripts (`analyses/`), the package they share (`src/mls_emergence/`, in which the copying model is `transmission/spatial.py`), the computed outputs (`output/`), the figures (`figures/`), and the tests (`tests/`). `MANIFEST.md` maps every output and figure to the script that writes it, and `run_all.sh` rebuilds the figures of the main text from the committed findings tables and then runs the analyses. Every number in this supplement names the script or output file it comes from.

The phase tests are also packaged as a tool, `phasecheck` (`src/phasecheck/`, guide `docs/PHASECHECK.md`), so that the questions asked here can be put to

another region or another phase scheme. It takes one table of assemblages, in a spreadsheet or a delimited file, with a name, a latitude and longitude, a phase label, and a count of sherds in each class per row, and it refuses tables it cannot use as counts (percentages, identifiers or totals treated as classes, phases with one member, study areas more than 1,000 km across). A plain run answers five questions in order: whether location alone recovers the phases (clustering on coordinates and composition at varying weights); whether the phase boundaries separate the assemblages better than same-size divisions around random centers and compact divisions (cultural $F_{ST}$ and the boundary excess, over posterior draws of the counts); whether each phase is distinct from the rest; whether each assemblage fits its own phase; and how different the phases are in sherds. With `--model` it calibrates the copying model with no groups to the record's diversity, screening 297 parameter combinations and confirming the matches on fresh runs, and answers four more: whether the model reproduces the record's diversity; whether the record differs between phases by more than the model produces at the confirmed combinations; whether simulated records sort into the phases; and whether the comparison could detect a restriction at the phase boundaries, from records simulated with and without one. It writes a report in Markdown with the tables behind it as CSV and JSON. A distance matrix along rivers or trails can replace straight-line distance for the boundary excess, and a column of dates or seriation scores can order the assemblages in time.

The tool implements the earlier version's tests, not the spatial field, latent-unit and barrier models or the diversity-matched neutral comparison of the main text. It is checked against the record of this study as the earlier version analyzed it (`examples/phasecheck/`, `tests/phasecheck/`). With the river distances and that version's seed it reproduces its phase-boundary comparison to the printed digit (between-phase $F_{ST}$ 0.0121; probabilities 0.57 and 0.72; boundary excess +7.3; probabilities 0.58 and 0.65), the recovery by location, each assemblage's fit, and the sherd differences; its copying model is the paper's model and, with the paper's seeds, gives the calibration's 51 matched and 48 confirmed combinations and the model comparison's 7 of 300 runs reaching the observed between-phase $F_{ST}$. It does not include local innovation as an alternative to a restriction, the map of compositional change, the settlement-gap test, or figures, and its calibration selects among the confirmed combinations by the difference between the phases rather than between the two spatial clusters. The probabilities it reports are the phase lines' rank among alternative lines, not significance levels, and on a record of this size the comparison discriminates only moderately, as its guide documents.

## References

Cobb, Charles R. 2000. *From Quarry to Cornfield: The Political Economy of Mississippian Hoe Production.* University of Alabama Press, Tuscaloosa.

Haley, Bryan S. 2014. The Big Picture at Hollywood: Geophysical and Ar-

chaeological Investigations at a Mississippian Mound Centre. *Archaeological Prospection* 21(1):39–47. https://doi.org/10.1002/arp.1477.

House, John H. 1993. Dating the Kent Phase. *Southeastern Archaeology* 12(1):21–32.

Lehner, Bernhard, and Günther Grill. 2013. Global River Hydrography and Network Routing: Baseline Data and New Approaches to Study the World's Large River Systems. *Hydrological Processes* 27(15):2171–2186.

Lipo, Carl P. 2001. *Science, Style and the Study of Community Structure: An Example from the Central Mississippi River Valley*. BAR International Series 918. Archaeopress, Oxford.

Mainfort, Jr., Robert C. 2001. The Late Prehistoric and Protohistoric Periods in the Central Mississippi Valley. In *Societies in Eclipse: Archaeology of the Eastern Woodlands Indians, a.d. 1400–1700*, edited by David S. Brose, C. Wesley Cowan, and Jr. Mainfort Robert C., pp. 173–190. Smithsonian Institution Press, Washington, DC.

Mainfort, Jr., Robert C. 2003. An Ordination Approach to Assessing Late Period Phases in the Central Mississippi Valley. *Southeastern Archaeology* 22(2):176–184.

Mitchem, Jeffrey M., David W. Stahle, Timothy S. Mulvihill, and Jami J. Lockhart. 2017. Making the Case for the Parkin Site as Casqui: Hernando de Soto's 1541 Cross.

Morse, Phyllis A. 1981. *Parkin: The 1978–1979 Archeological Investigations of a Cross County, Arkansas Site*. Arkansas Archeological Survey Research Series 13. Arkansas Archeological Survey, Fayetteville.

Morse, Phyllis A. 1990. The Parkin Site and the Parkin Phase. In *Towns and Temples Along the Mississippi*, edited by David H. Dye and Cheryl Anne Cox, pp. 118–134. University of Alabama Press, Tuscaloosa.

Phillips, Philip. 1970. *Archaeological Survey in the Lower Yazoo Basin, Mississippi, 1949–1955*, Vol. 60. Papers of the Peabody Museum of Archaeology and Ethnology. Peabody Museum, Harvard University, Cambridge, MA.

Reimer, Paula J. et al. 2020. The IntCal20 Northern Hemisphere Radiocarbon Age Calibration Curve (0–55 Cal kBP). *Radiocarbon* 62(4):725–757. https://doi.org/10.1017/RDC.2020.41.

Smith, Gerald P. 1990. The Walls Phase and Its Neighbors. In *Towns and Temples Along the Mississippi*, edited by David H. Dye and Cheryl Anne

Cox, pp. 135–169. University of Alabama Press, Tuscaloosa.

Surovell, Todd A., and P. Jeffrey Brantingham. 2007. A Note on the Use of Temporal Frequency Distributions in Studies of Prehistoric Demography. *Journal of Archaeological Science* 34(11):1868–1877. https://doi.org/10.1016/j.jas.2007.01.003.